

REVIEW

Biotransformation Effect of *Bombyx Mori* L. May Play An Important Role in Treating Diabetic Nephropathy*

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ABSTRACT Compared with herbal drugs, medicine processed from animals (animal medicine) was thought to have more bioactive substances and higher activities. Biotransformation effect often plays an important role in their effect. However, researches about effect of animal medicine on diabetic nephropathy and applying animal medicine as natural bio-transformer were seldom reported. The purpose of this paper was to reveal the use of *Bombyx Mori* L. on diabetic nephropathy from ancient to modern times. The classical literature indicated that Saosi Decoction (缲丝汤), which contains *Bombyx Mori* L. or silkworm cocoon, was applied to treat disorders congruent with modern disease diabetic nephropathy from the Ming to Qing Dynasty in ancient China. Modern studies showed that *Bombyx Mori* L. contains four main active constituents. Among these, 1-deoxynojirimycin (1-DNJ) and quercetin showed promising potential to be new agents in diabetic nephropathy treatment. The concentrations of 1-DNJ and the activities of quercetin in *Bombyx Mori* L. are higher than in mulberry leaves, because of the biotransformation in the *Bombyx Mori* L. body. However, these specific components need further human and mechanistic studies to determine their therapeutic potential for this challenging condition.

KEYWORDS diabetic nephropathy, *Bombyx Mori* L., 1-deoxynojirimycin, quercetin, biotransformation, Chinese medicine

Diabetic nephropathy is the most common cause of end-stage renal disease around the world and is characterized by fast progress and a poor prognosis.⁽¹⁾ In China, after thousands of years of traditional medical practice, a great deal of valuable experience has accumulated regarding diabetic nephropathy. Effect of herbal medicine in the treatment of diabetic nephropathy has been confirmed by systematic reviews, including *Astragalus Membranaceus*,⁽²⁾ *Ginkgo Biloba* extract,⁽³⁾ *ligustrazine*⁽⁴⁾ and so on. However, the therapeutic effect of Chinese medicine (CM) was based on its bioactive component. But herbal medicine often contains too little active ingredients. For example, level of paclitaxel, harringtonine, camptothecin and ginsenoside Rh2 contained in plants were less than 1/10000.⁽⁵⁾

Biotransformation was a metabolic process that exogenous chemical substance alternated by or in a biological system, resulting in new bioactive compound. The aim of biotransformation of CM was to promote treatment effect, enhance bioavailability and improve universality of CM preparation. Currently, artificial method such as microbial transformation and enzymatic conversion was the mainstream in biotransformation technique.^(6,7) But researches applying animal medicine as natural bio-transformer

were seldom reported. Besides, animal medicine itself also contains many active agents. Some researches indicated that animal medicine was more particularly useful in improving microcirculation than the herbal medicine,^(8,9) and some of the animal medicines were effective in reducing proteinuria even used solely.⁽⁹⁻¹³⁾ However, researches about effect of animal medicine on diabetic nephropathy were rare.

Our previous work⁽¹⁴⁾ identified the classical terms of diabetic nephropathy in ancient China. Classical terms associated with diabetic nephropathy were collected through original research articles,

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the empirical experience of venerable CM doctors, medical monographs and textbooks. A nationwide expert consultation was done to obtain experts' opinion on which terms were more correspond with diabetic nephropathy. These classical terms identified by expert consultation were included in the further verification. Their corresponding modern diseases were retrieved in the textbooks and monographs of Chinese internal medicine, monographs of Shen (Kidney) disease of CM, and dictionaries of CM. As a result, Xiaoxiao (下消), Shexiao (肾消) and Xiaoshen (消肾) were considered more correspond with diabetic nephropathy.

Our study searched ancient records of Xiaoxiao, Shexiao and Xiaoshen via "Encyclopedia of Traditional Chinese Medicine." Chinese formulae for treating disorders congruent with diabetic nephropathy recorded in medical books from Tang Dynasty (618 AD to 907 AD) to Qing Dynasty (1644 AD to 1912 AD) were collected. Among these ancient formulae, Saosi Decoction (缲丝汤) was the simplest since it only contained one ingredient – *Bombyx Mori* L. and/or silk cocoon. Through modern literature review, we found that the biotransformation effect of *Bombyx Mori* L. had an important potential in treating diabetic nephropathy.

The purpose of this paper was to reveal the use of *Bombyx Mori* L. on diabetic nephropathy from ancient to modern times. Active ingredients and biotransformation effect of *Bombyx Mori* L. in treatment of diabetic nephrology were introduced.

Ancient Records about *Bombyx Mori* L. in Diabetic Nephropathy Treatment

Bombyx Mori L. was recorded in ancient literatures for treating diabetic nephropathy as a single-item recipe.

The earliest record of its use was in *Medicine Biography* (Yi Xue Zheng Zhuan) written by YU Chuan during the Ming Dynasty (1515 AD). In reference to the inherited formula Yuancanjan Decoction (原蚕茧汤, also called Saosi Decoction, he wrote that: 'it had an excellent effect on Shexiao with the symptoms of turbid urine, polydipsia and excessive appetite but the person loses weight... the effect of the hot water used in reeling silk (i.e. Saosi Decoction) is best. If this is not available, it can be replaced by a decoction of silkworm cocoon or silk floss'.—from *Medicine Biography*.⁽¹⁵⁾

From the Ming to Qing Dynasty, there were 9 ancient books recording the use of *Bombyx Mori* L. on treating diabetic nephropathy. The single-item recipe was called Yuancanjan Decoction, or Caosi Decoction (缲丝汤), or Saosi Decoction or Danjiansi Decoction (单茧丝汤). These ancient records all confirmed the effect of *Bombyx Mori* L. on classical disease Xiaoxiao, Shexiao, and Xiaoshen, which have the similar pathogenesis and clinical manifestation as modern disease diabetic nephropathy.

Active Ingredients of Saosi Decoction and Their Effect on Diabetic Nephropathy

No modern studies of 'Yuancanjan Decoction', 'Caosi Decoction', 'Saosi Decoction' or 'Danjiansi Decoction' were located but there has been considerable research into *Bombyx Mori* L., its related products, and its active ingredients. According to modern studies, the *Bombyx Mori* L. and its products are rich in various active substances such as alkaloids, flavonoids, and silk protein hydrolysates.

Alkaloids

1-deoxynojirimycin (DNJ) is a major component of the alkaloids in silkworm.⁽¹⁶⁾ The N-hydroxyethyl derivative of 1-DNJ, miglitol, has been marketed as an antidiabetic drug and was approved by the Food and Drug Administration in 1996.⁽¹⁷⁾ Yokoyama H, et al⁽¹⁸⁾ showed that miglitol decreased the urinary albumin excretion rate in Japanese patients with type 2 diabetes. One possible mechanism is related to improved insulin resistance,⁽¹⁹⁾ which was associated with podocyte injury mediated by nucleotide-binding oligomerization domain containing 2.⁽²⁰⁾ Uzu T, et al⁽²¹⁾ showed that miglitol could reduce serum interleukin (IL)-18 levels in patients with stage 3 diabetic nephropathy. Elevated serum IL-18 level was associated with the increased urinary protein excretion and the decreased glomerular filtration rate.⁽²²⁻²⁴⁾ Serum IL-18 is a proinflammatory cytokine which contributed to upregulation of intercellular cell adhesion molecule-1⁽²⁵⁾ and interferon- γ ,⁽²⁶⁾ increased production of other proinflammatory cytokines [IL-1 and tumor necrosis factor- α],⁽²⁷⁾ and apoptosis of endothelial cells.⁽²⁸⁾ All these inflammatory cytokines participated in the development of diabetic nephropathy.⁽²⁹⁾ They played an important role in kidney fibrosis through activating fibroblast in diabetes, such as epithelial-to-mesenchymal transition and endothelial-to-mesenchymal transition (EndMT).⁽³⁰⁾ Also they were

involved in proliferation of mesangial cells and matrix synthesis, glomerular basement membrane (GBM) thickening, increased fibronectin expression, and enhanced endothelial permeability,⁽³¹⁾ resulting in progressive fibrosis in the diabetic kidney.^(32,33)

Although miglitol is not recommended for patients with renal insufficiency (serum creatinine >2 mg/dL) because it is excreted primarily via the kidney.⁽¹⁸⁾ It was reported to be safe for patients with stage 3 diabetic nephropathy, reducing blood glucose without kidney function deterioration.⁽³⁴⁾

Four other alkaloids [fagomine, 3-epi-fagomine, 1,4-dideoxy-1,4-imino-D-arabinitol, and 1,4-dideoxy-1,4-imino-(2-O-b-D-glucopyranosyl)-D-arabinitol] have also been reported from silkworm.⁽³⁵⁾ Fagomine showed an anti-hyperglycemic effect by potentiating glucose-induced insulin secretion in healthy rats.⁽³⁶⁾ However, its effect on diabetic nephropathy has not been reported.

Flavonoids

Flavonoids, which have been purified and identified from the sericin layer of silkworm cocoons, were shown to decrease glucose level by α -glucosidase inhibition, free-radical scavenging, and antioxidation.⁽³⁷⁾ According to reported studies, two C-prolinylquercetins (6-C-[(2S,5S)-prolin-5-yl] quercetin and 6-C-[(2S,5R)-prolin-5-yl] quercetin),⁽³⁸⁾ three kaempferol glycosides (Kaempferol 5-O-glucoside, kaempferol 5,4'-di-O-glucoside, and Kaempferol 7-O-glucoside),^(39,40) and eight quercetin glycosides (quercetin 5-O-glucoside, quercetin 7-O-glucoside, quercetin 4'-O-glucoside, quercetin 5,4'-di-O-glucoside, quercetin 3,7-di-O-glucoside, quercetin 3,3'-di-O-glucoside, quercetin 5,3'-di-O-glucoside, and quercetin 5,7,4'-tri-O-glucoside),^(39,41) have also been isolated from silkworm cocoon.

Among these flavonoids, quercetin was reported to have renal protective effects. It suppressed glomerular mesangial cell hypertrophy, proliferation, and extracellular matrix accumulation, as all of which occur in glomerular sclerosis.⁽⁴²⁾ Proposed mechanisms of action include inhibition of transforming growth factor- β 1 (TGF- β 1) expression⁽⁴⁵⁾ and amelioration of oxidative stress,^(44,45) which have been shown to be final common mediators of renal injury in diabetes.⁽⁴⁶⁾ Additionally, quercetin was reported to reduce nuclear factor- κ B expression, which may be

involved in the pathogenesis of proteinuria in diabetic nephropathy.^(47,48)

Silk Protein Hydrolysates

Fibroin hydrolysates from silkworm cocoons or whole cocoon hydrolysates containing both fibroin and sericin were reported to have hypoglycemic effects in *in vivo* studies.⁽⁴⁹⁻⁵¹⁾ They increased serum insulin level and improved the sensitivity to insulin by upregulating the expression of insulin-like growth factor-1 (IGF-1),⁽⁵²⁾ glucose transporter-1 (GLUT-1),^(49,53) and GLUT-4.⁽⁴⁹⁾ Jung, et al⁽⁵¹⁾ in Korea classified silk protein hydrolysates as an insulin secretagogue, which has actions similar to sulfonylureas and meglitinides. Although, their effects on diabetic nephropathy have not been reported, one study in Korea showed that silk fibroin exhibited anti-inflammatory activity in mice edema model.⁽⁵⁴⁾

The active substances sequestered by *Bombyx Mori* L. in their bodies or cocoons are derived from their only food, mulberry leaves, which have been used in modern medicine as dietary supplements to treat diabetes and its complications.⁽⁵⁵⁾ This raised the question: What is the difference between the effect of silkworm or silk cocoon and mulberry leaves on diabetic nephropathy?

The Biotransformation Effect of *Bombyx Mori* L.

The *Bombyx Mori* L. alkaloids can also be isolated from mulberry leaves, the host plant of the silkworm,⁽⁵⁶⁾ but the amounts of alkaloids differ between them. For example, it has been reported that the 1-DNJ content in silkworms is 2.7-fold that in mulberry leaves.⁽³⁶⁾ The mechanism of DNJ accumulation in silkworms may be the metabolic conversion of 2-O-a-D-galactopyranosyl-DNJ (GAL-DNJ) to DNJ by α -galactosidase and/or β -galactosidase⁽⁵⁷⁾ (Figure 1). In addition, mulberry latex is exuded by mulberry trees for plant defence against insect herbivory. The concentrations of 1-DNJ in mulberry latex are higher than in whole mulberry leaves using traditional extraction methods with solvents.⁽⁵⁸⁾ The silkworm, *Bombyx Mori* L., has evolved enzymatic adaptation to the toxicity of mulberry latex,⁽⁵⁹⁾ so higher concentrations of 1-DNJ may be accumulated in its body after feeding.

Flavonoids isolated from silkworm cocoons are structurally different from those in mulberry

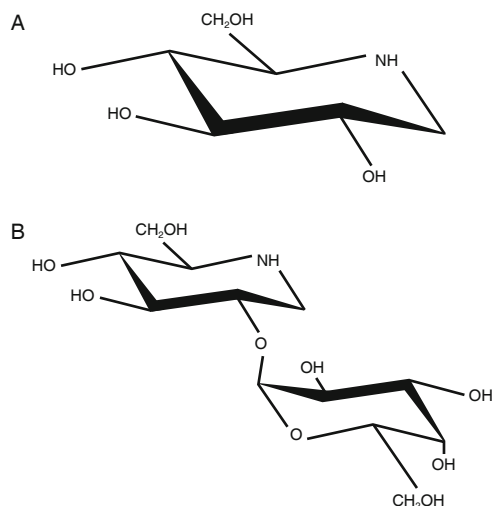


Figure 1. Chemical Structures of DNJ (A), GAL-DNJ (B)

leaves, and it is thought that dietary flavonoids are biotransformed in silkworms. For example, quercetin 5-O-glucoside, quercetin 5,4'-di-O-glucoside, and quercetin 5,7,4'-tri-O-glucoside⁽³⁹⁾ (Figure 2A) extracted from silkworm cocoons were formed from dietary quercetin in mulberry leaves by the activity of UDP-glucosyltransferase enzyme in silkworms⁽⁶⁰⁾ Prolinalin A (Figure 2B) and B (Figure 2C) are produced from dietary quercetin (Figure 2D) due to not only the glycosylation reactions, but also the dehydration reaction with L-5-hydroxyproline.⁽³⁸⁾

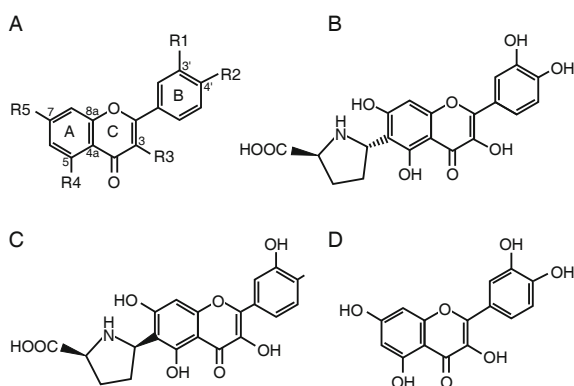


Figure 2. Chemical Structures of Dietary Quercetin and Quercetin in Silkworm Cocoon

Notes: A: quercetin 5-O-glucoside: R1=R2=R3=R5=OH, R4=O-Glucose; quercetin 5,4'-di-O-glucoside: R1=R3=R5=OH, R2=R4=O-glucose; quercetin 5,7,4'-tri-O-glucoside: R1=R3=R5=OH, R2=R4=O-glucose; B: prolinalin A; C: prolinalin B; D: dietary quercetin

Additionally, the bioconjugation of insulin with silk fibroin⁽⁶¹⁾ or silk sericin⁽⁶²⁾ greatly improved its physicochemical and biological stability, thereby enhancing the hypoglycemic effect. With the aid of

modern technology, the silkworm has been used as a "bioreactor" to obtain active insulin fusion protein conjugated with cholera toxin B subunit, which was able to induce insulin-specific oral tolerance in diabetes treatment.⁽⁶³⁻⁶⁶⁾ The properties that allow the silkworm to function as an expression system or "bioreactor" may account for the difference between the effect of silkworm and mulberry leaves.

Modern Biosynthesis of Active Ingredients

Multiple agents working synergistically in one formula is a major feature of CM. As the human body is an extremely complex network, the 'one-disease-one drug' paradigm often fail to achieve therapeutic effect. Researchers began to promote a new drug discovery paradigm "combining existing active ingredients".

Scholars in China have reported notable suppressive effects of the compound prescription of 1-DNJ and natural flavonoids on high blood glucose levels.⁽⁶⁷⁾ A new patented antidiabetic drug consisting of *Bombyx Mori* L. excrement and *Radix Glycyrrhizae*, called Jintangning Capsule (金糖宁胶囊), had a similar hypoglycemic effect to acarbose in phase 3 clinical trials in China.⁽⁶⁸⁾ 1-DNJ is the main component of silkworm excrement and *Radix Glycyrrhizae* is rich in active flavonoids. Nevertheless, there are no *in vitro* or *in vivo* studies describing the effect of the combined use of 1-DNJ and flavonoids on diabetic nephropathy.

Discussion and Conclusions

DNJ, which was first isolated from the mulberry tree by Yagi, et al in 1976, accounted for 50% of the mulberry iminosugars.⁽³⁵⁾ However, the DNJ content of commercial mulberry products can be as low as 0.1%,⁽⁶⁹⁾ which is insufficient for an effective dose in humans.⁽⁷⁰⁾ DNJ-enriched products are desirable. Thus, the accumulation of DNJ in the *Bombyx Mori* L. may be helpful for developing a new drug based on DNJ. Studies in our literature review showed that DNJ could decrease the urinary albumin excretion rate and serum IL-18 levels in patients with diabetic nephropathy.

The possible mechanisms were related to improving insulin resistance and anti-inflammation. However, the strength of the clinical evidences for a renal protective effect is limited by small numbers of patients in the studies. More researches regarding the effect of DNJ on diabetic nephropathy and related

mechanisms are needed. Moreover, because of its renal excretion,⁽⁷¹⁾ the safety of 1-DNJ in patients with renal insufficiency must be assessed.

Quercetin, a well-known bioflavonoid, is present in many foods such as onions, apples, broccoli, green tea, and red wine. At present, it is widely used as an effective antioxidant or a commercial food additive in the food processing industry.⁽⁷²⁾ Studies have shown that quercetin has a definite renal protective effect, with antioxidant activity and α glucosidase inhibition.⁽⁷³⁻⁷⁵⁾ Our review found that dietary quercetin biotransformed in the silkworm had greater biological activity than quercetin from mulberry. Thus, quercetin from the silkworm may have potential as a new agent for diabetic nephropathy. However, most of the studies describing the effect of quercetin on diabetic nephropathy were based on *in vitro* and *in vivo* experiments. Studies in humans are required.

Silk protein has been used as a food additive since the 1990s, as it is harmless to humans.⁽⁷⁶⁾ It has hypoglycemic effects via the upregulation of IGF-1, GLUT-1, and GLUT-4 expression. However, the persistent accumulation of IGF-1 in renal tissue may play an important role in the progression of diabetic nephropathy by decreasing extracellular matrix degradation⁽⁷⁴⁾ and stimulating mesangial cell proliferation.⁽⁷⁷⁻⁸⁰⁾ GLUT-1 may also participate in the progression of diabetic nephropathy by facilitating the production of TGF- β 1 and the formation of advanced glycosylation end-products and their receptors.⁽⁸¹⁾ At present, the inconsistent findings cannot be explained, and the effects of silk protein hydrolysates on diabetic nephropathy are uncertain.

In conclusion, *Bombyx Mori* L. has promising potential for development as new agent for the treatment of diabetic nephropathy. The main mechanism includes anti-inflammation, antioxidant activity and improved insulin resistance effect. Higher concentrations of active agents in *Bombyx Mori* L. than in mulberry leaves, as a result of the biotransformation in the silkworm body, would be helpful in its effect on diabetic nephropathy. But more researches regarding the effect of *Bombyx Mori* L. or its active agents on diabetic nephropathy and related mechanisms were needed.

Moreover, we should still pay attention to the

side effects of animal medicine. Foreign protein contained in animal medicine may cause some allergenic symptoms in patients such as skin pruritus or more severe allergenic reactions. So these specific components need further human and mechanistic studies to determine their therapeutic potential for this challenging condition.

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